



The Limits of Temporal Prediction in Short-Form Video: Transformer Sequence Modeling and Evidence for Variable Reinforcement on TikTok

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Abstract

This thesis investigates whether temporal Transformer models can predict watch time behavior on TikTok using only publicly available user data. I trained a custom Transformer encoder on sequential video interaction data from three user case studies spanning seven years of TikTok usage, with input features limited to watch time history, latency, and a binary "like" indicator. The results reveal a three-group gradient in predictive capability. For low-frequency users, no model — including the Transformer — could outperform a naïve mean baseline, suggesting that sporadic engagement is fundamentally stochastic. For medium-frequency users, the Transformer achieved only marginal improvement over baseline, consistent with variable reinforcement patterns that resist temporal prediction. For high-frequency users, both the Transformer and linear regression modestly outperformed the baseline, indicating that heavy usage produces measurable behavioral regularity detectable from temporal features alone. Two additional findings emerged: the binary "like" feature had no significant effect on watch time prediction, suggesting that traditional engagement signals no longer drive scrolling behavior on TikTok, and the inclusion of second-of-day as an input feature caused the model to overfit to daily routines rather than learn genuine sequential dependencies. These findings suggest that TikTok's recommendation algorithm operates primarily through variable reinforcement rather than engagement-based feedback loops.

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1 Introduction

1.1 Motivation

The majority of existing research into social media addiction relies on survey-based methods or conventional classifiers applied to aggregate engagement features. While these approaches have established that addiction is measurable and classifiable, they do not model the sequential dynamics of how users actually interact with content over time. This is a significant gap. TikTok’s recommendation algorithm is widely understood to operate in part due to variable reinforcement — a mechanism analogous to slot machines — in which content quality is deliberately inconsistent to sustain engagement through unpredictability. If this is the case, then the temporal sequence of a user’s interactions should contain detectable patterns that reflect the algorithm’s influence on behavior. My motivation was to test this directly: rather than surveying users about their habits, I sought to model the behavioral trace data itself and determine whether a Transformer architecture designed specifically for sequence modelling could learn the temporal dependencies underlying scrolling habits. The practical implications are substantial. If temporal features alone can predict watch time as a proxy for engagement intensity, this opens a pathway toward early detection tools that do not require invasive data collection or self-reported surveys. Conversely, if the model fails to predict behavior, that failure itself is informative. It suggests that the algorithm’s variable reinforcement strategy is effective enough to resist temporal prediction, which carries its own implications for platform regulation and user autonomy.

1.2 Contributions

This thesis makes four primary contributions. First, I designed and trained a custom temporal Transformer encoder on sequential TikTok video interaction data, using only three publicly available features (watch time history, latency, and a “like” indicator) to predict future watch time behavior at the individual user level. Second, through three longitudinal case studies spanning seven years of TikTok usage, I identify a three-group behavioral gradient in which predictability itself scales with usage intensity: low-frequency users exhibit stochastic behavior that no model can outperform a naïve baseline on, medium-frequency users resist prediction in a manner consistent with variable reinforcement, and high-frequency users display measurable behavioral regularity detectable from temporal features alone. Third, I provide empirical evidence against engagement-based addiction models by demonstrating that the “like” feature has no significant effect on binge prediction, suggesting that TikTok has shifted away from traditional engagement signals toward content-driven variable reinforcement. Fourth, I identify and document how second-of-day as an input feature causes the model to overfit to natural daily routines rather than learn genuine sequential dependencies, a methodological finding relevant to future behavioral time-series research.

2 Background and Related Work

2.1 Social Media Addiction and User Behavior

The rapid rise of short-form video platforms such as TikTok, Instagram Reels, and Douyin has prompted growing research into behavioral addiction as a public health concern. The focus of my research centers around TikTok specifically. As of February 2026, the app has 1.9 billion global monthly active users, with the US composing 136 million of those users (SQ Magazine, 2026). The majority of the users fall into younger demographics with adults ages 18-29 representing 63% of users. In addition, 68% of teens ages 13-17 report using the platform with 1 in 5 U.S. teens report being on TikTok “almost constantly” (Pew Research Center, 2026). With these high rates of usage, researchers have sought explanations into how to better characterize social media addiction and resulting attention-decay. Ye et al., (2025) constructed a model exploring the relationships between short video usage intensity, perceived mood enhancement (termed “TikTok Brain”), short video addiction, and decreased attention control among Chinese adolescents with an average age of 19.84. Their work employed traditional statistical analysis rather than machine learning, identifying significant pathways between usage intensity and addictive behaviors. Similarly, Silalahi, (2026) assessed social media addiction levels among university students focused on Instagram and TikTok, employing a Naive Bayes classifier to categorize users into low, moderate, and high addiction levels, achieving 90% accuracy on TikTok data. While these studies establish that social media addiction is measurable and classifiable, they rely on survey-based or simple probabilistic methods rather than modeling the temporal dynamics of user behavior directly.

Most closely related to my work, Yang et al., (2025) combined survey responses with donated digital trace data from 1,590 TikTok users to study behavioral addiction. They stratified users into three addiction groups and trained classifier models using basic engagement features, achieving an F1 score of approximately 0.55 for identifying highly addicted users. Notably, their findings suggest that predicting addictive behavior solely from usage data is “rather difficult,” a conclusion my own results corroborate. However, their approach used conventional classifiers on aggregate features rather than modeling the sequential structure of user interactions over time.

2.2 Transformer Architectures and Attention Mechanisms

The Transformer architecture, introduced by Vaswani et al., (2017), established a new paradigm for sequence modeling based entirely on self-attention mechanisms. The core innovation replaces recurrence and convolutions with scaled dot-product attention, defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where Q , K , and V represent query, key, and value matrices respectively. The original Transformer employs multi-head attention and position-wise feed-forward networks within each encoder layer, connected through residual connections and layer normalization. This

architecture demonstrated superior performance on machine translation tasks while being significantly more parallelizable than recurrent alternatives.

Since its introduction, the Transformer has been adapted across diverse domains. Wang et al., (2023) extended the Transformer to multi-modal fake news detection, demonstrating the architecture’s flexibility in processing heterogeneous input types through their Multi-Modal Transformer using Two-level Visual Features (MTTV). In sequential recommendation, Kang and McAuley, (2018) proposed SASRec, a self-attentive model that captures long-term user behavior semantics while making predictions based on relatively recent actions, outperforming both Markov Chain and RNN-based sequential models. Their work is particularly relevant to this thesis, as it demonstrates the Transformer’s effectiveness in modeling sequential user interaction sequences.

3 Methodology

3.1 Data Collection and Feature Definitions

In adherence to the Human Research Protection Program (HRPP) at the University of Virginia, I complied with federally mandated research guidelines for non-medical behavioral human subjects research U.S. Department of Health and Human Services, (2018). A central methodological constraint of this study was the deliberate use of only publicly available user data. Under data portability regulations such as the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA), social media platforms are required to provide users with access to their own data in a structured, machine-readable format. TikTok complies with these regulations through an in-app data download tool that allows any user to request an export of their interaction history. Critically, the data provided through this tool constitutes observed data, records of what the user did, rather than inferred data, such as the behavioral profiles or interest models that TikTok constructs internally. This means that the algorithmic features TikTok uses to curate each user’s feed, including content embeddings, predicted engagement scores, and recommendation weights, remain inaccessible.

This constraint was intentional. The goal of this study was to evaluate how much of TikTok’s addictive behavioral patterning can be detected and predicted using only the data available to users themselves. TikTok’s recommendation algorithm is largely opaque meaning a black box whose internal mechanisms are proprietary and undisclosed. By restricting the input feature space to what users can actually access, this study tests whether publicly available signals are sufficient to surface the behavioral signatures of algorithmic conditioning. If temporal features alone can predict binge behavior, it suggests that the algorithm’s influence leaves detectable traces in observable data, opening a pathway toward algorithmic interpretability that does not require proprietary access. Conversely, if prediction fails, it implies that the algorithm’s mechanisms are sufficiently obscured to resist external analysis from user-side data alone. This finding presents its own implications for transparency and platform accountability.

I collected data from 10 individuals using TikTok’s data download tool. From this initial pool, I selected three individuals ($N = 3$) representing distinct points along the usage intensity spectrum: a low-frequency outlier, a moderate-frequency user representative

of typical engagement, and an extreme high-frequency user. This selection strategy was designed to capture the full range of behavioral patterns rather than focusing exclusively on heavy usage. I explicitly sought to design my research as a case study given the large datasets (specifically regarding the medium and high-frequency users) and the longitudinal value of 7 years. Support for this type of small-N design is emerging within the human-computer interaction community with Smith and Little, (2018) arguing that if psychology is to be a mature quantitative science, then its primary theoretical aim should be to investigate systematic, functional relationships as they manifest themselves at the individual participant level.

The three users engaged with the app from June 2019 to February 2026, representing approximately seven years of usage. The resulting longitudinal dataset comprised 141,553 discrete video interaction sequences across the three users, providing the temporal depth required to train a time-series Transformer. Participants were 20–22-year-old college students at the time of data collection, a demographic representing the primary cohort on TikTok, with 63% of U.S. adults aged 18–29 reporting platform use (Pew Research Center, 2026).

The feature space was categorized into three distinct sets: temporal, engagement, and system features.

- **Temporal Features:** These consisted of the raw timestamp for each interaction. For the modeling process, these were decomposed into the date and time of day to capture both long-term trends and circadian patterns.
- **Engagement Feature:** This included binary indicators for active user interactions, specifically *likes* extracted from the user’s activity logs.
- **System Features:** These were programmatically derived from the sequential timestamp data to approximate internal platform metrics. To maintain data integrity, two specific filters were applied during this derivation: a lower-bound filter of 1.5 seconds was implemented to remove “micro-scrolls” (accidental swipes), and an upper-bound cap of 180 seconds was used to exclude instances of idle phone noise or app exits.
 - **Estimated Watch Time (W):** This represents the duration of a viewing event, calculated as the total seconds elapsed between video n and video $n + 1$.
 - **Latency (L):** Defined as the delay between the user’s intent to scroll and the subsequent video successfully loading. In the absence of proprietary server-side logs, latency was approximated as the residual time within the interaction interval not accounted for by active playback.

I also implemented a *Session ID* feature using a 30-minute inactivity threshold. If the gap between two video interactions exceeded 1,800 seconds, the sequence was marked as the start of a new session. This allowed the Transformer to reset its positional encodings and avoid learning dependencies across disjointed periods of use.

3.2 Model Architecture

I formulated the watch time prediction task as a sequence-to-scalar regression problem. Given a sequence of the k most recent video interactions, each represented as a feature

vector $\mathbf{x}_i \in \mathbb{R}^3$ consisting of historical watch time, latency, and a binary "like" indicator, the model predicts the expected watch time $\hat{y} \in \mathbb{R}_{\geq 0}$ for the next video. I fix $k = 20$, representing the user's twenty most recent interactions. I initially included a fourth input feature for my model "Second of Day" which represented the exact time of day in which they watched each video in the sequence. However, primary results utilizing Shapley values demonstrated that the model was relying disproportionately on second of day whose natural bias was positively skewing my results.

3.2.1 Preprocessing

Prior to model input, I apply a $\log(1+x)$ transformation to the historical watch time feature to compress the right-skewed distribution and mitigate the influence of extreme outliers, some exceeding 35,000 seconds prior to filtering. The continuous features, watch time and latency, are then standardized using z-score normalization:

$$x' = \frac{x - \mu}{\sigma} \quad (2)$$

where μ and σ are the mean and standard deviation computed over the training set. The binary like feature is left unscaled to preserve its interpretability as a categorical trigger. The target variable undergoes the same $\log(1+x)$ transformation, with the inverse transform $\text{expm1}(\hat{y}) = e^{\hat{y}} - 1$ applied at inference to recover predictions on the original scale.

3.2.2 Encoder Architecture

The core of my model is a custom Transformer encoder. Each input feature vector $\mathbf{x}_i \in \mathbb{R}^3$ is first projected to a higher-dimensional representation $\mathbf{h}_i \in \mathbb{R}^{128}$ through a learned linear embedding:

$$\mathbf{h}_i = \mathbf{W}_{\text{proj}} \mathbf{x}_i + \mathbf{b}_{\text{proj}}, \quad \mathbf{W}_{\text{proj}} \in \mathbb{R}^{128 \times 3}, \quad \mathbf{b}_{\text{proj}} \in \mathbb{R}^{128} \quad (3)$$

I selected a model dimension of $d_{\text{model}} = 128$ to provide sufficient representational capacity while mitigating overfitting risk given the low dimensionality of the input features. The projected representations are augmented with positional encodings to inject sequential ordering information into the model, as the self-attention mechanism is otherwise permutation-invariant.

The encoded sequence is then passed through my stacked Transformer encoder layers ($N = 4$). Each layer consists of a single-head self-attention sublayer followed by a position-wise feedforward sublayer. The self-attention operation is defined as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} \right) \mathbf{V} \quad (4)$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are the query, key, and value matrices respectively, and d_k is the dimension of the key vectors. For sequences shorter than k , inputs are left-padded with zeros, and padded positions are masked from the attention softmax by setting their pre-softmax scores to $-\infty$, ensuring they do not contribute to the attended representation.

Each transformer block follows the Pre-Norm convention (Xiong et al., 2020), in which layer normalization is applied to the sublayer input rather than its output:

$$h' = h + \text{Sublayer}(\text{LayerNorm}(h)) \quad (5)$$

This contrasts with the original Post-Norm formulation of Vaswani et al., (2017) and has been shown to improve gradient flow and training stability in deep transformers, removing the need for learning-rate warmup. A final layer normalization is applied to the output of the stack before pooling.

The self-attention mechanism within each layer allows every position in the sequence to attend to every other position, enabling the model to capture dependencies between interactions regardless of their temporal distance within the window. This is particularly relevant for my task, as a user’s engagement pattern on a video eight interactions ago may still be informative of current watch behavior.

3.2.3 Output Head

Following the final encoder layer, I aggregated the sequence representation using masked mean pooling over all non-padded positions. Given the output matrix $\mathbf{H}^{(N)} \in \mathbb{R}^{k \times 128}$ and a binary mask $\mathbf{m} \in \{0, 1\}^k$ indicating valid (non-padded) positions, the aggregated representation is:

$$\bar{\mathbf{h}} = \frac{\sum_{i=1}^k m_i \cdot \mathbf{h}_i^{(N)}}{\sum_{i=1}^k m_i} \quad (6)$$

This pooling strategy allows the model to incorporate information from all observed interactions in the sequence rather than relying solely on the most recent position, while ignoring padded positions for users with fewer than k interactions. The aggregated vector is then passed through a two-layer feedforward network to produce a scalar prediction:

$$\hat{y} = \text{ReLU}(\mathbf{w}_2^\top \text{ReLU}(\mathbf{W}_1 \bar{\mathbf{h}} + \mathbf{b}_1) + b_2), \quad \mathbf{W}_1 \in \mathbb{R}^{32 \times 128}, \quad \mathbf{w}_2 \in \mathbb{R}^{32} \quad (7)$$

The final ReLU activation constrains the output of the log-space regression to be non-negative, which, combined with the `expm1` inverse transform at inference, guarantees non-negative watch time predictions

4 Results

4.1 SHAP Analysis and the Second-of-Day Bias

My initial model results included the Second of Day feature, however due to the bias as seen below, these results were ruled inconclusive. I utilized SHAP framework (SHapley Additive exPlanations), where each feature is assigned an importance value for a particular prediction (Lundberg and Lee, 2017). It is also important to note that I trained the majority of model on the medium-frequency case data as it was more representative of the average TikTok user.

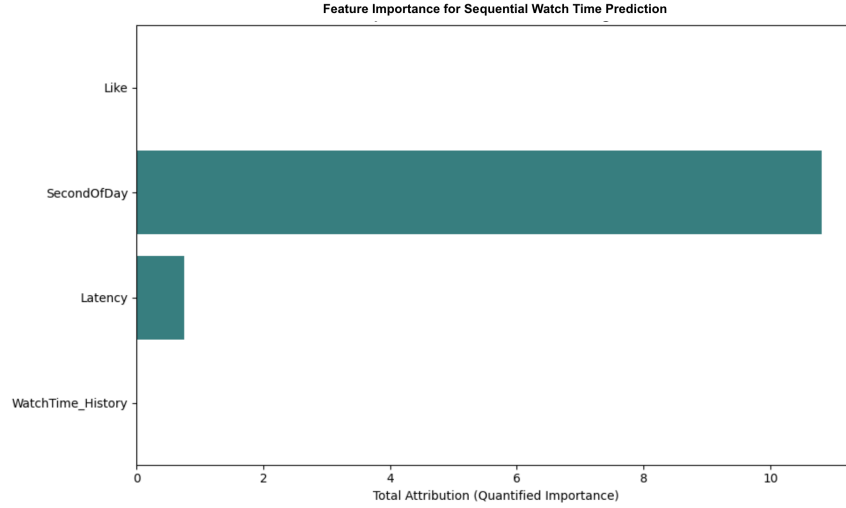


Figure 1: SHAP feature importance for the initial model including Second of Day. Second of Day dominates attribution, obscuring behavioral signal.

Following these SHAP results, I returned to my preprocessing script and changed it accordingly. With this change, the bias within my model was removed as seen below:

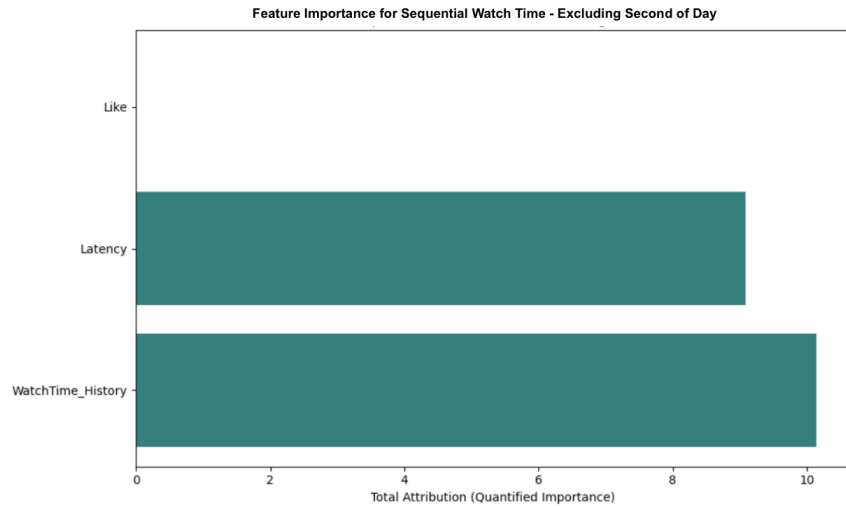


Figure 2: SHAP feature importance after removing Second of Day. Watch Time, History and Latency emerge as primary contributors. Like has negligible attribution.

This resulted in my naive baseline improving from -69.24% to -2.95%. This significant improvement in variance meant the model no longer relying on easier temporal shortcuts but rather leaning towards behavioral learning. The temporal shortcuts can be better defined as the model simply learning that users are more likely to scroll at say a second of day that is later at night instead of in the middle of the day where the individual may have work

or class. This ties into past related work on the ties between social media and night-time use with 70% of people report using social media after getting into bed (Sleep Foundation, 2025). Out of curiosity, I ran a simple statistical analysis with the medium-frequency user’s raw csv data to prove this hypothesis. Out of the 33,802 data points, approximately 50.1% fell between 9 pm and 6 am. See Table 1 below for more details.

Table 1: TikTok Usage Distribution by Time Block

Time Block	Videos	%
Night (9 PM–6 AM)	16,918	50.1
Afternoon (12 PM–5 PM)	7,979	23.6
Evening (5 PM–9 PM)	6,533	19.3
Morning (6 AM–12 PM)	2,372	7.0
Total	33,802	100.0

To optimize the model’s capacity, I conducted a hyperparameter search using a held-out validation set. This resulted in the selection of a 4-layer encoder architecture and a reduced learning rate of 0.0001 (from 0.0005) to ensure stable gradient descent. These final configurations yielded a 1.61% improvement over the Naive Baseline. Interestingly, the SHAP feature importance measures changed with WatchTime History proving to be roughly 1/3 more indicative of TikTok Watch Time Prediction than latency. The third feature importance, the binary ”like” feature, did not serve as a mechanism of binge predicting behavior as seen in the SHAP figures above.

4.2 Sequence Length

Following this, I sought to experiment to see if sequencing length affected user prediction and if I could better fine-tune this aspect. Table 2 presents the effect of sequence length on model performance for medium users. Increasing the sequence length from 10 to 20 timesteps yielded a marginal improvement across all metrics for the Transformer model, reducing MAE from 14.55s to 14.52s and RMSE from 36.63s to 36.45s. Notably, the mean baseline remained identical across both configurations (14.58s MAE), confirming that the improvement is attributable to the model capturing longer-range temporal dependencies rather than a difference in data distribution. While the improvement is modest, it suggests that a longer historical window provides additional — albeit limited — sequential signal for the Transformer’s attention mechanism. Based on this finding, a sequence length of 20 timesteps was adopted as the standard configuration for all subsequent experiments across user groups. I completed separate timestep testing however, due to the smaller dataset of the low usage case, and marginal improvements, I kept the 20 timestep sequence length.

Table 2: Effect of sequence length on medium user predictions

Sequence Length	Model	MAE (s)	RMSE (s)
10 timesteps	Mean Baseline	14.58	36.58
	Linear Reg.	14.53	36.47
	Transformer	14.55	36.63
20 timesteps	Mean Baseline	14.58	36.58
	Linear Reg.	14.54	36.43
	Transformer	14.52	36.45

4.3 Cross-Group Model Comparison

Table 3 presents model performance across all three user groups using a sequence length of 20 timesteps.

The Sporadic User: The sporadic-user group contained only 230 training sequences, reflecting the sparse and irregular engagement patterns of casual TikTok users. Linear regression produced unstable estimates due to insufficient training data (MAE: 170.65s, RMSE: 601.69s), indicating that no stable linear relationship exists between temporal features and watch time for this group. The Transformer performed comparably to the mean baseline (35.16s vs 34.37s MAE) but could not outperform it. This suggests that low-user behavior is fundamentally stochastic — there is insufficient temporal structure for any model to learn from. Additionally, the high baseline MAE of 34.37s reflects extreme variance in watch time, consistent with users who engage unpredictably and without established viewing habits.

Moderate User: With substantially more data (27,040 training sequences), all three models converged to near-identical performance (MAE range: 14.52–14.58s). The Transformer achieved the lowest MAE (14.52s) and RMSE (36.45s), marginally outperforming both baselines. However, the differences are negligible, suggesting that while medium users generate sufficient data to stabilize predictions, the underlying engagement pattern remains largely governed by variability. Notably, the 0.06-second MAE improvement of the Transformer over the naïve mean baseline is not statistically significant, falling well within the margin of test-set variance. This null result is itself substantively informative as it indicates that the Transformer’s self-attention mechanism cannot extract meaningful sequential dependencies from medium-frequency viewing behavior. This outcome aligns with variable reinforcement ie: the deliberate unpredictability of content quality prevents any temporal model from reliably anticipating the next watch duration.

High-Frequency User: The high-frequency-user group contained 114,283 training sequences and produced the most differentiated results. Both the Transformer and linear regression outperformed the mean baseline (MAE: 7.97s and 7.95s vs 8.11s respectively). Linear regression achieved the lowest MAE, while the Transformer achieved the lowest RMSE (17.79s vs 17.92s), suggesting it better captures extreme values through sequential pattern recognition. Crucially, the baseline MAE of 8.11s is substantially lower than the medium (14.58s) and low (34.37s) groups, indicating that high-user watch times are inherently more consistent — their behavior has become habitual rather than stochastic.

Table 3: Model performance across user groups (sequence length = 20)

User Group	Model	MAE (s)	RMSE (s)
Low (n=230)	Mean Baseline	34.37	55.85
	Linear Reg.	170.65	601.69
	Transformer	35.16	56.40
Medium (n=27,040)	Mean Baseline	14.58	36.58
	Linear Reg.	14.54	36.43
	Transformer	14.52	36.45
High (n=114,283)	Mean Baseline	8.11	18.23
	Linear Reg.	7.95	17.92
	Transformer	7.97	17.79

4.4 Summary of Findings

As usage intensity increases, baseline variability decreases (MAE: 34.37s $\rightarrow$ 14.58s $\rightarrow$ 8.11s) and temporal features become increasingly predictive. For low-frequency users, no model can outperform a simple mean. For medium users, marginal gains are possible but negligible. For high-frequency users, temporal features, particularly watch time history and latency, provide genuine predictive signal. This progression suggests that TikTok’s algorithmic conditioning produces measurable behavioral regularity in heavy users, and that this regularity is detectable using only publicly available temporal features.

5 Discussion

The findings of this study suggest a fundamental shift in the relationship between the user and the platform as usage intensity scales.

5.1 Predictability Ceiling and Algorithmic Capture

The most compelling finding of this research is the collapse of behavioral variance in Case 3. For the high-frequency user, these results are consistent with a near-linear temporal process. This indicates a transition from Content-Based Addiction, where the user seeks specific videos to, Process-Based Addiction. The user is now synchronized with the temporal flow of the scroll itself. I would also like to acknowledge an alternative interpretation, that is low variance (not linearity) is why linear-regression and the transformer converge.

5.2 Redefining Short-Form Video as Social Media

These results raise a broader question: Is short-form video content still appropriately categorized as social media? The definition of social media, as outlined by Wolfers et al., (2025), describes it as an umbrella concept encompassing social software and social networking. However, short-form video content as it currently operates may no longer fit comfortably within this definition.

Unlike platforms such as Facebook or Instagram, TikTok users tend to spend the majority of their time on the algorithmically curated "For You" page rather than the socially

driven "Friends" page. The finding that the binary "like" feature had limited effect on watch time prediction supports this: users are less motivated by videos they actively enjoy, but rather by the random possibility of a dopamine response. In short, I believe the "social" dimension is debatable. TikTok functions more as a personalized, algorithmically optimized medium than a social network.

5.3 Second of Day Bias and Digital Routine

The initial model's reliance on the "Second of Day" feature aligns with research from the Sleep Foundation (2025), which notes that 70% of social media users engage with platforms after getting into bed. While this feature initially inflated the model's accuracy, its removal was necessary to isolate genuine sequential dependencies. This highlights a methodological challenge in behavioral time-series research: distinguishing between a user's habitual daily schedule and the platform's specific behavioral conditioning.

5.4 Limitations

I, of course, need to acknowledge the significant limitations within my research. The first being the small participant pool of $N = 3$ and the niche participant demographic of 20-22 year olds. This emphasizes that my findings are an exploratory case study rather than representing all users. Additionally the component of self-selection bias in participant recruitment is important to note. Second, my manipulation of time variables. My watch time is estimated from inter-event latency, not the exact server-side data. It also assumes that the user was fully engaged on their phone for the entire watch time, where more realistically, they could have their phone running or be engaged in other tasks as multitasking while using social media is a known phenomenon. However, I addressed this through my elimination of outliers in my pre-processing data stage. Another limitation is the lack of content features in the input vector. I did consider exploring the impact of "hashtagging" on predicting user watch time, however, decided to not include that in my final model. This was in part due to reasons that popped up during my literature review of the topic. Specifically the development of hashtagging from the initial date of the data collected (2019) to the final date (2026). Introduced by Twitter (now x) in 2007, the hashtag became a tool to boost content visibility and track engagement and analytics (Reeves, 2025). However, in recent years post widespread AI adoption, specifically within TikTok, it is no longer a sufficient tool for attention in comparison to other ingredients. Thus, this feature inclusion would have opened up an entirely other research question of the impact of hashtags on modern viewing behavior, while equally interesting, is not what I intended to explore.

6 Conclusion

Through three longitudinal case studies spanning seven years, this study sought to answer whether temporal Transformer models could predict watch time behavior on TikTok. The results reveal a three-group behavioral gradient:

Case 1 (Sporadic): Behavior remains fundamentally stochastic and human-centric, resisting all forms of temporal prediction.

Case 2 (Moderate): Behavior mirrors the variable reinforcement of a "slot machine," where marginal predictive gains are possible but content remains the primary driver.

Case 3 (Chronic): Behavior becomes highly deterministic. With 114,283 interactions, this user displayed a "temporal rhythm" so regular that the Transformer achieved a 7.97s MAE, outperforming the mean baseline. Additionally the transformer achieved slightly better success in handling outliers (RMSE).

These findings provide empirical evidence that as addiction scales, the triggers designed to increase engagement (like variable reinforcement) are no longer the primary cause of continued use. Instead, the algorithm's conditioning produces a behavioral regularity that renders the user predictable.

6.1 Future Work and Speculative Reflections

Future research should expand the participant pool and incorporate hashtag data to determine if content categories can further bridge the gap between linear and Transformer models. On a more speculative note, this research invites reflection on the trajectory of social media. While algorithmic strategies have been optimized for addiction, there may be an upper bound to this optimization. Human impatience and neurochemical oversaturation may eventually provide a natural expiration point for the variable reinforcement model. Speaking from my own experience as a high-frequency user of TikTok, I no longer get the same dopamine reward from using the app. The algorithm has become over-optimized for increased scrolling, now showing me content that I enjoy far less. I find myself using the app at a lower rate, and if this is at all true for the rest of the population, I think we are in good hands regarding social media addiction. My hope is that social media will eventually return to its original purpose: a mechanism for social networking and community, rather than algorithmically optimized passive consumption.

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